Microsoft Defender for Endpoint Integration

Implementation Solution Guide

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Executive Summary — Microsoft Defender for Endpoint Integration (Microsoft Intune)

White-label · Intune Solutions

Antivirus alone doesn't stop a modern attacker. **Microsoft Defender for Endpoint (MDE)** adds endpoint detection and response (EDR), behavioral protection, and a live device **risk score** — and when integrated with Microsoft Intune, devices onboard automatically and that risk score flows straight into compliance and Conditional Access.

This is a complete, **deploy-it-today** package for integrating Microsoft Defender for Endpoint with Microsoft Intune, written for an administrator with no prior MDE experience.

What you get

- A plain-language **Solution Overview** of how the Intune ■ MDE integration works.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click steps in the required order.
- Five **PowerShell scripts** for onboarding discovery, sensor validation, tenant reporting, remediation, and connector status.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

The **service-to-service connector** between Intune and MDE, an **EDR onboarding policy** that silently onboards enrolled devices, and the **machine-risk-score signal** wired into device compliance — so a compromised device automatically loses access.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Full EDR coverage with automatic onboarding and a risk score that powers Zero Trust enforcement — deployable by a single administrator and validated before fleet rollout.

1. Solution Overview — Microsoft Defender for Endpoint Integration

What this integration is

Microsoft Defender for Endpoint (MDE) is Microsoft's enterprise endpoint security platform — next-generation antivirus, **endpoint detection and response (EDR)**, attack surface reduction, and

threat/vulnerability management. This solution connects MDE to **Microsoft Intune** so Intune **onboards** devices to MDE automatically and consumes MDE's **device risk score** as a compliance and Conditional Access signal.

How it works

1. You enable the **service-to-service connection** between Intune and the Defender portal.
2. An Intune **EDR onboarding policy** carries the onboarding configuration to enrolled devices, which begin reporting to MDE.
3. MDE continuously evaluates each device and assigns a **machine risk score** (Low/Medium/High).
4. Intune **device compliance** can require a risk score at or below a threshold; a risky device becomes **Not compliant**.
5. **Conditional Access** (require compliant device) then blocks the risky device until it's remediated.

What you can manage through the integration

- **EDR onboarding** of Windows devices via Intune (no manual onboarding scripts at scale).
- **Risk-based compliance** — fail compliance when MDE risk exceeds your threshold.
- A foundation for managing **Defender Antivirus**, **Attack Surface Reduction**, and **firewall** policy from Intune (companion solutions).

Benefits

- **Automatic onboarding** — enrolled devices join MDE with no per-device scripting.
- **Risk-driven access** — compromised devices automatically lose access via compliance + CA.
- **Single pane** — manage endpoint security policy from Intune.
- **Faster response** — EDR detection, investigation, and automated remediation.

Limitations (be honest with buyers)

- Requires **Defender for Endpoint licensing** (P1/P2, M365 E5/E5 Security, or Defender for Business).
- The risk-score → access flow needs the **Device Compliance** and **Conditional Access** solutions.
- Onboarding takes effect after policy sync and a device check-in to the Defender service.
- Down-level/unsupported OSes have reduced capability.

The standard deployment model

> ***Enable the connector → create the EDR onboarding policy → assign to a pilot → wire risk score into compliance → validate → expand.*** *repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Microsoft Defender for Endpoint Integration

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Defender for Endpoint licensing

Description: Microsoft Defender for Endpoint P1/P2, Microsoft 365 E5 / E5 Security, or Defender for Business. **Why:** MDE onboarding and EDR require an MDE plan. **Verify:** M365 admin center → **Billing** → **Licenses**; the Defender portal (`security.microsoft.com`) loads. **Implement:** Assign M365 E5/E5 Security, standalone Defender for Endpoint, or Defender for Business (with Business Premium).

2.2 Intune + enrollment

Description: Intune subscription with target devices enrolled. **Why:** Intune delivers the onboarding policy and reads the risk score. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.3 Access to the Defender portal

Description: Security Administrator access to `https://security.microsoft.com`. **Why:** The Intune connection is enabled from the Defender portal settings. **Verify:** The portal loads; **Settings** → **Endpoints** → **Advanced features** is accessible. **Implement:** Assign **Security Administrator** (or Global Admin for setup).

2.4 Supported Windows OS

Description: Windows 10/11 (Defender sensor built in). **Why:** Onboarding and EDR require a supported, patched Windows. **Verify:** `winver`; the **Sense** service exists (`Get-Service Sense`). **Implement:** Patch/upgrade to a supported build.

2.5 Network (Defender endpoints)

Description: Outbound HTTPS to Defender for Endpoint service URLs. **Why:** The sensor reports telemetry and receives configuration from the service. **Verify:** From a device, the MDE connectivity test passes (or run `MDE-Validation.ps1`). **Implement:** Allow Microsoft Defender for Endpoint URLs/IPs through proxy/firewall; do not SSL-inspect MDE traffic.

2.6 Compliance + Conditional Access (for risk enforcement)

Description: Device compliance and Conditional Access in place. **Why:** To turn the MDE risk score into access enforcement. **Verify:** Intune → **Devices** → **Compliance**; Entra → **Conditional Access**. **Implement:** Deploy the **Device Compliance** and **Conditional Access** solutions.

Checklist

- Defender for Endpoint licensing assigned
- Devices enrolled in Intune
- Defender portal access (Security Administrator)
- Windows 10/11 with the Sense service — `MDE-Discovery.ps1`
- Network allows MDE endpoints (no SSL inspection)
- (For enforcement) Compliance + Conditional Access solutions in place
- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Microsoft Defender for Endpoint Integration

Follow these steps **in order**. You use both the **Defender portal** (`security.microsoft.com`) and the **Intune admin center** (`intune.microsoft.com`).

> ***Model:*** *Enable the connector → EDR onboarding policy → assign to pilot → wire risk into compliance.*

Step 1 — Turn on the Intune connection (Defender portal)

1. Open `https://security.microsoft.com` → **Settings** → **Endpoints** → **Advanced features**.
2. Turn **Microsoft Intune connection** to **On**. **Save preferences**.

Step 2 — Confirm the connector in Intune

1. Intune → **Endpoint security** → **Microsoft Defender for Endpoint**.
2. Confirm **Connection status: Available / Active** (wait a few minutes / **Refresh** if needed).
3. Set **Allow Microsoft Defender for Endpoint to enforce Endpoint Security Configurations** to **On** if you'll manage security config from Intune.

Step 3 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `MDE-Pilot` → add pilot devices/users. **Create**.

Step 4 — Create the EDR onboarding policy

1. Intune → **Endpoint security** → **Endpoint detection and response** → **Create Policy**.

2. **Platform:** Windows. **Profile:** Endpoint detection and response. **Create**.

3. **Name:** `MDE - EDR Onboarding - Pilot`. **Next**.

4. **Microsoft Defender for Endpoint client configuration package type: Auto from connector** (the onboarding blob is supplied by the connection). **Sample sharing:** as per policy. **Next**.

Step 5 — Assign the onboarding policy

1. **Assignments** → **Add groups** → `MDE-Pilot`. **Next** → **Create**.

Step 6 — Wire the risk score into compliance

1. Intune → **Devices** → **Compliance** → edit your Windows compliance policy (from the Device Compliance solution).

2. Under **Microsoft Defender for Endpoint**, set **Require the device to be at or under the machine risk score** to **Medium** (or **Low** for stricter scopes). **Save**.

Step 7 — (Recommended) enforce with Conditional Access

1. Ensure a Conditional Access policy **requires a compliant device** (Conditional Access solution). A device exceeding the risk threshold becomes Not compliant and loses access.

Step 8 — Validate the pilot

1. On a pilot device, run `MDE-Discovery.ps1` and `MDE-Validation.ps1` → confirm onboarded + sensor healthy.

2. Run `MDE-ConnectorCheck.ps1` → confirms the Intune↔MDE connector is Active.

3. Run `MDE-TenantReport.ps1` → confirms devices appear onboarded with a risk score. Complete the **Testing Guide**.

Step 9 — Expand to production

1. Add a production group to the onboarding policy **Assignments**; roll out in waves.

2. Confirm risk-based compliance behaves as expected before broad Conditional Access enforcement.

3. Re-validate each wave with `MDE-Validation.ps1` / `MDE-TenantReport.ps1`.

Order of operations

`Enable connector → Confirm in Intune → Pilot group → EDR onboarding policy → Assign → Risk into compliance → (CA require compliant) → Validate → Expand`

5. Testing Guide — Microsoft Defender for Endpoint Integration

Run after the pilot onboarding policy applies (Implementation Step 8) and after each production wave.

Test 1 — Connector active

Do: Intune → **Endpoint security** → **Microsoft Defender for Endpoint**; or run `MDE-ConnectorCheck.ps1`. **Expected:** Connection status **Active / Available**.

Test 2 — Onboarding policy applied

Do: Intune → **Endpoint security** → **Endpoint detection and response** → policy → **Device status**. **Expected:** Pilot devices show **Succeeded**.

Test 3 — Device onboarded (local)

Do: On a pilot device run `MDE-Validation.ps1` (checks the Sense service + onboarding state). **Expected:** Sense service **Running**; `OnboardingState = 1`.

Test 4 — Device appears in the Defender portal

Do: `https://security.microsoft.com` → **Assets** → **Devices**. **Expected:** The pilot device is listed with health **Active** and a risk level.

Test 5 — Risk score feeds compliance

Do: Intune → device → **Compliance**; review the Defender risk requirement. **Expected:** A device under the threshold is **Compliant**; a higher-risk device would be **Not compliant**.

Test 6 — EDR detection (optional, safe simulation)

Do: Run Microsoft's official EDR detection test (from the Defender portal device page) on a test device. **Expected:** An alert appears in the Defender portal, confirming EDR telemetry.

Test 7 — Tenant report

Do: Run `MDE-TenantReport.ps1`. **Expected:** Onboarded devices with health + risk score listed.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–5 pass for the pilot group
Wave pass	Tests 2–3, 5, 7 pass before the next wave

6. Troubleshooting Guide — Microsoft Defender for Endpoint Integration

Use `MDE-Validation.ps1` / `MDE-Discovery.ps1` for local onboarding state and `MDE-ConnectorCheck.ps1` for the connector, `MDE-Remediation.ps1` for safe fixes.

A. Connector problems

Symptom	Cause	Remediation
Connector "Unavailable" in Intune	Intune connection not turned on in Defender portal	Defender portal → **Settings** → **Endpoints** → **Advanced features** → **Microsoft Intune connection = On**
Connector active but no onboarding	No onboarding policy / wrong package type	Create the EDR onboarding policy with **Auto from connector**

B. Device not onboarding

Symptom	Cause	Remediation
Policy "Succeeded" but device absent in Defender	Sensor not reporting yet, or network blocked	Run `MDE-Validation.ps1`; allow MDE URLs; reboot; wait for first check-in
`OnboardingState = 0`	Onboarding not applied	Sync policy (`MDE-Remediation.ps1`); confirm device in `MDE-Pilot` group
Sense service stopped	Service disabled/crashed	`MDE-Remediation.ps1` restarts Sense; check Event Viewer (SENSE)

C. Risk score not affecting compliance

- Compliance unchanged by risk → the compliance policy's Defender risk requirement isn't set, or the connector isn't active. Set **Require risk at or under <level>** in the compliance policy.
- Device risk "Unknown" → not yet onboarded/reporting; wait for first telemetry.

D. Network / connectivity

- Sensor unhealthy → run the MDE connectivity analyzer; ensure MDE service URLs are allowed and **not SSL-inspected**.
- Proxy issues → configure the sensor proxy (static or WinHTTP) per Microsoft guidance.

E. Diagnostics quick reference

```
Get-Service Sense                                    # Defender for Endpoint sensor service
Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\Windows Advanced Threat Protection\Status'  # OnboardingState
Get-MpComputerStatus                                 # Defender AV health
```

Sensor logs/connectivity: the **MDEClientAnalyzer** tool from Microsoft.

F. When to escalate

If the device shows `OnboardingState = 1`, Sense is Running, and connectivity passes but the device never appears in the Defender portal, capture the device ID and open a Microsoft Defender support case.

7. Best Practices — Microsoft Defender for Endpoint Integration

- **Enable the connector first.** Nothing onboards through Intune until the Defender↔Intune connection is On and Active.
- **Use Auto-from-connector onboarding.** Let the connector supply the onboarding blob — no manual scripts to manage at scale.
- **Pilot before fleet.** Confirm onboarding, sensor health, and portal visibility on a small group first.
- **Start the risk threshold at Medium.** It catches real risk without over-blocking; tighten to Low for sensitive scopes.
- **Layer compliance + Conditional Access.** The risk score only enforces access when wired into compliance and a CA "require compliant device" policy.
- **Don't SSL-inspect MDE traffic.** It breaks the sensor; exclude Defender URLs from inspection.
- **Watch sensor health.** Track onboarded/health/risk with `MDE-TenantReport.ps1`; chase inactive sensors.
- **Run the EDR detection test.** Validate telemetry end-to-end with Microsoft's official detection test on a pilot device.
- **Manage AV/ASR/firewall through Intune** once onboarded (companion solutions) for a single policy plane.
- **Document the risk policy.** Record the threshold and rationale for audits.

8. Deliverables Summary — Microsoft Defender for Endpoint Integration

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What the Intune ■ MDE integration is, how it works, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once

Document	Purpose
3. Implementation Guide	Exact, ordered steps across the Defender portal and Intune
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`MDE-Discovery.ps1`	Local onboarding readiness: Sense service, OS, network to MDE
`MDE-Validation.ps1`	Confirms the device is onboarded (OnboardingState=1) and the sensor is healthy
`MDE-TenantReport.ps1`	Tenant-wide onboarded devices with health + risk score (Microsoft Graph)
`MDE-Remediation.ps1`	Safe remediation; restarts Sense, forces sync, runs a quick scan
`MDE-ConnectorCheck.ps1`	Confirms the Intune ↔ Defender connector is Active (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Full EDR coverage with automatic onboarding and a risk score that powers Zero Trust enforcement.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

MDE-ConnectorCheck.ps1

MDE-Discovery.ps1

MDE-Remediation.ps1

MDE-TenantReport.ps1

MDE-Validation.ps1